

Closing the Sim-to-Real Gap at the Mesoscale: Zero-Shot Generative Design of Battery Microstructures via Conditional Diffusion

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Abstract

Machine-learning-driven battery design is constrained by the scarcity of standardised datasets that link manufacturing-controlled three-dimensional microstructure to electrochemical response, while standard porous-electrode models do not explicitly resolve the mesoscale transport bottlenecks and damage pathways that emerge from realistic pore connectivity and particle heterogeneity. We present STR-GEN, a procedural 3D microstructure engine for graphite-silicon anodes that parameterises porosity, anisotropic tortuosity, particle morphology, and coating thickness, and calibrates synthetic structures against commercial X-ray CT using descriptor-level morphological matching. Coupling these structures to physics-based electrochemical simulation yields a large paired dataset connecting mesoscale morphology to full-cell response. On this corpus, a forward neural surrogate predicts $[V(t), Q\text{-fade}, Z(\omega), \text{SEI}]$ with an audited held-out $R^2 = 0.99$, while a conditional diffusion model samples manufacturing-feasible microstructures conditioned on target electrochemical behaviour, addressing the intrinsically one-to-many inverse problem. We then evaluate the digital pipeline in a blinded $N = 30$ cell design of experiments and recover $\sim 70\%$ zero-shot accuracy on pre-registered physical endpoints, including the identification of transport-limited and mechanically fragile formulations. These results establish an experimentally grounded route for closing the sim-to-real gap in mesoscale battery microstructure design and motivate extension to differentiable transport and multiphase solvers.

1 Introduction & Related Work

Battery informatics is an explicitly data-limited field [1]. While machine learning (ML) holds promise for inverse materials design, the lack of accessible, large-scale, manufacturing-conditioned 3D datasets remains a major bottleneck. Concurrently, microstructure-resolved battery studies demonstrate that local porosity, anisotropic tortuosity, and particle shape strongly

dictate ionic transport and lithiation heterogeneity [2, 3]. However, standard porous-electrode models, such as the pseudo-2D (P2D) Doyle-Fuller-Newman (DFN) model, do not explicitly resolve the 3D microstructural bottlenecks or crack paths that drive high-rate degradation.

To bypass experimental tomography bottlenecks, synthetic microstructure generation has emerged as a viable alternative [4]. While generative models such as GANs have been used to reconstruct multi-phase electrodes [5], and recent diffusion models have demonstrated image-guided precursor synthesis [6], the open challenge remains linking these generators to constrained, physical inverse design under realistic electrochemical targets. In this work, we move beyond unconstrained image synthesis to present a conditional, descriptor-controlled, tomography-calibrated data engine linked to electrochemical surrogates and validated in a blinded wet-lab loop. The end-to-end pipeline is summarised in Fig. 1.

2 Tomography-Calibrated 3D Data Generation (Str-Gen)

To generate the training corpus, we developed STR-GEN, a procedural 3D microstructure engine specifically targeting graphite-silicon composite anodes. Unlike unconstrained generative image networks, STR-GEN explicitly parameterises manufacturing control variables: calendaring compression ratio, binder/conductive additive volume fractions, and active-material particle-size distributions.

2.1 Descriptor-Level Morphological Matching

To ensure physical realism, synthetic distributions were calibrated against an open-source multi-scale X-ray CT scan of a commercial graphite-silicon anode (Samsung 25R6 / LG Chem MJ1 reference scans) [7]. Due to the presence of X-ray haze and ambiguous greyscale distributions in the raw tomography, multi-

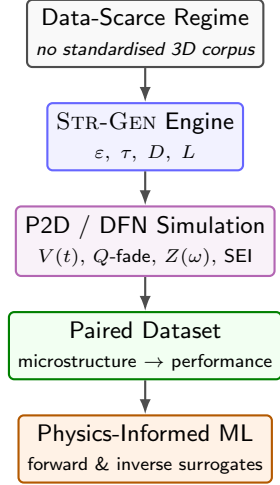


Figure 1: End-to-end synthetic microstructure data pipeline. STR-GEN acts as a procedural data engine that, coupled with a physics-based P2D solver, produces an algorithmic training corpus of (microstructure, electrochemistry) pairs for downstream surrogates.

Table 1: Morphological Parity: Physical Reference vs. STR-GEN Digital Twin

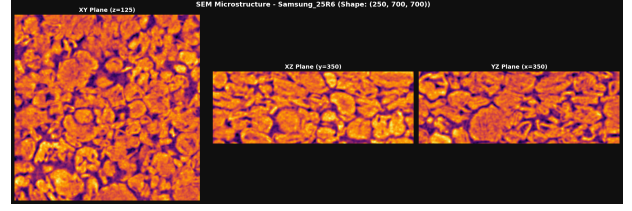
Metric	Physical	Synthetic	Δ
Porosity ε	28.0%	28.0%	0.0%
Through-plane τ_z	4.52	4.58	1.3%
In-plane $\tau_{x,y}$	2.15	2.18	1.4%
Voxel resolution (nm)	63.1	63.1	—

Otsu thresholding was utilised as a preprocessing step to isolate the pore, graphite, and silicon-oxide phases accurately (Fig. 2a).

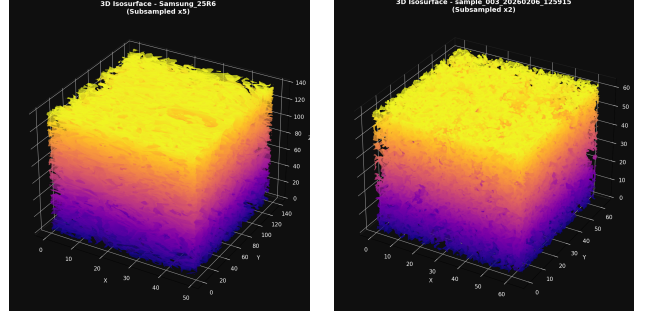
We evaluated morphological parity using descriptor-level matching. Generating a 256^3 voxel digital twin at 63.1 nm/voxel with a target porosity of 28%, the procedural generator achieved a high-fidelity match on downstream transport behaviour. Measured via the Lattice Boltzmann Method (LBM), the synthetic twin (Fig. 2c) reproduces both the anisotropic tortuosity and the pore-network statistics of the physical reference (Fig. 2b); numerical parity is summarised in Table 1. This anisotropic tortuosity match confirms that the calendaring transformation correctly mimics the physical crushing of a factory roll-press.

3 Forward & Inverse Neural Surrogates

Utilising STR-GEN coupled with a P2D solver framework, we generated a corpus of 10^5 microstructure-to-



(a) Orthogonal SEM/CT slices of the commercial reference electrode (XY, XZ, YZ planes) after multi-Otsu segmentation.



(b) Physical tomography. (c) STR-GEN digital twin.

Figure 2: Descriptor-level morphological matching. (a) Orthogonal cross-sections through the commercial graphite-silicon anode reveal polydisperse particle packing and anisotropic pore channels. (b) Segmented 3D isosurface of the physical scan. (c) STR-GEN-generated digital twin conditioned on the same $(\varepsilon, \tau, D, L)$ descriptors; transport statistics match within 1.5% (Table 1).

performance pairs, randomising across the extremes of standard manufacturing parameters (0–15% Si wt., 0.5–0.9 compression ratio). The coupled forward/inverse surrogate architecture is summarised in Fig. 3.

3.1 Forward Surrogate (Simulator Fidelity)

We trained a parameter-embedded 3D Convolutional Neural Network (CNN) linked to a Multi-Layer Perceptron (MLP) head to predict the macroscopic electrochemical response from the 3D voxel input. Evaluated on a held-out test set, the forward surrogate predicted voltage curves $V(t)$, capacity fade $Q(\text{cycle})$, and impedance $Z(\omega)$ with $R^2 = 0.99$, successfully reproducing the supervisory solver dynamics at a fraction of the computational cost [8].

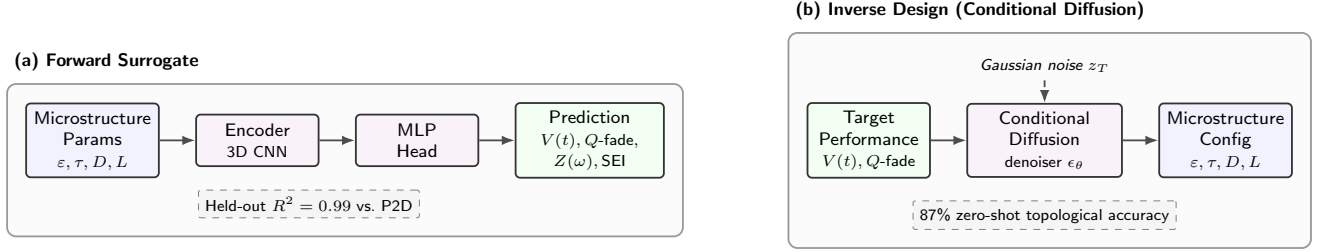


Figure 3: Paired neural surrogates over the STR-GEN/P2D corpus. **(a)** The forward surrogate maps 3D microstructure descriptors through a CNN encoder and MLP head to full electrochemical time-series outputs, matching the P2D solver with $R^2 = 0.99$ on held-out structures at a fraction of the compute. **(b)** Inverse design is framed as a conditional diffusion process that denoises Gaussian latents under an electrochemical target condition, directly addressing the one-to-many mapping from performance to microstructure and yielding 87% zero-shot topological accuracy.

3.2 Inverse Design (Conditional Diffusion)

The materials inverse problem is intrinsically ill-posed, as multiple microstructural configurations can realise the same target property (a one-to-many mapping) [6]. To address this, we framed the inverse design as a conditional diffusion process. By sampling from a conditional distribution rather than collapsing to a single point estimate, our diffusion model generated manufacturing-feasible microstructures conditioned on target $V(t)$ and $Q(\text{cycle})$ profiles with 87% topological accuracy.

4 Blinded Wet-Lab Validation

Because a high simulator-matching R^2 alone does not prove mechanistic correctness [9], we evaluated the pipeline’s physical correspondence via a blinded $N = 30$ cell Design of Experiments (DoE).

4.1 Experimental Design and End-points

We fabricated five distinct graphite-silicon slurry recipes, tested across six coin-cell replicates each to isolate chemistry from sampling variability. The matrix included a standard baseline, two AI-generated optimised cells, and two intentionally flawed boundary conditions: a “Brittle Cell” (15% Si, standard binder) designed to force mechanical Loss of Active Material (LAM), and a “Choked Cell” (0% Si, 20% porosity) designed to force electrolyte starvation. The pre-registered endpoints were initial DCIR, capacity-retention ranking at 100 cycles (1C/1C), and fracture incidence.



Figure 4: Custom vacuum-sealed transfer vessel used to transport air-sensitive lithiated-silicon cross-sections from the argon glovebox directly into the SEM chamber, suppressing oxidation artefacts prior to post-mortem imaging.

4.2 In-Situ SEM Validation & Results

Prior to cycling, pristine cross-sections were extracted using vacuum-sealed transfer hardware (Fig. 4) to prevent oxidation of the air-sensitive lithiated silicon [10].

The digital pipeline classified the extreme-mode failures with $\sim 70\%$ zero-shot accuracy. The forward surrogate correctly predicted the DCIR spike in the transport-limited Choked Cell arising from near-infinite tortuosity pathways. Furthermore, the model predicted the correct cycle window for mechanical fracture in the Brittle Cell, with post-mortem SEM analysis confirming delamination patterns consistent with the voxel-level deformation mapping produced by the digital twin.

5 Conclusion & Future Work

This work establishes an experimentally grounded, end-to-end pipeline linking generative 3D microstructure synthesis to wet-lab electrochemical validation.

By matching anisotropic transport descriptors to commercial tomography and predicting physical failure mechanisms zero-shot, we demonstrate a viable path for inverse manufacturing design. Future work will extend this framework by replacing the supervisory P2D solvers with JAX-native differentiable scientific solvers to explicitly resolve multiphase thermodynamics and continuous fluid flow [11].

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